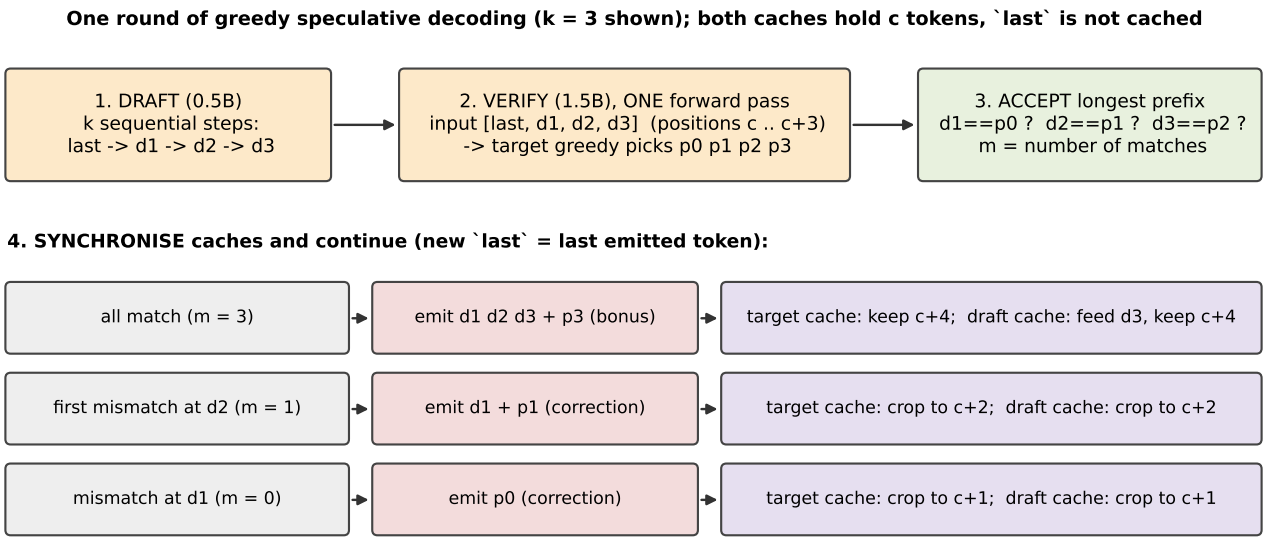
One round of greedy speculative decoding (k = 3 shown); both caches hold c tokens, `last` is not cached
1. DRAFT (0.5B)
k sequential steps:
last -> d1 -> d2 -> d3
2. VERIFY (1.5B), ONE forward pass
input [last, d1, d2, d3] (positions c .. c+3)
-> target greedy picks p0 p1 p2 p3
3. ACCEPT longest prefix
d1==p0 ? d2==p1 ? d3==p2 ?
m = number of matches
4. SYNCHRONISE caches and continue (new `last` = last emitted token):
all match (m = 3)
emit d1 d2 d3 + p3 (bonus)
target cache: keep c+4; draft cache: feed d3, keep c+4
first mismatch at d2 (m = 1)
emit d1 + p1 (correction)
target cache: crop to c+2; draft cache: crop to c+2
mismatch at d1 (m = 0)
emit p0 (correction)
target cache: crop to c+1; draft cache: crop to c+1
Every emitted token is the target model's own greedy choice -> output identical to plain greedy decoding; 1 to k+1 tokens per target pass.